

C++ File Handling — Quick Summary Sheet

File handling in C++ allows permanent data storage using files (.txt, .dat, etc.). The <fstream> header provides three main classes.

Class	Purpose
ofstream	Write to a file (Output File Stream)
ifstream	Read from a file (Input File Stream)
fstream	Both read and write (File Stream)

File Opening Modes

Mode	Description
ios::in	Open for reading
ios::out	Open for writing (overwrites existing data)
ios::app	Append new data at the end of file
ios::ate	Open and move write pointer to end
ios::binary	Open file in binary mode
ios::trunc	Delete old contents if file exists

Basic Examples

Write to file:

```
ofstream file("data.txt");
file << "Hello";
file.close();
```

Read from file:

```
ifstream file("data.txt");
string line;
while(getline(file, line)) cout << line << endl;
file.close();
```

Append mode:

```
ofstream file("data.txt", ios::app);
file << "New line";
```

Common Mistakes

Mistake	Correction
Forgetting to close file	Always call .close()
Overwriting old data	Use ios::app to append
Reading without checking existence	Use if(!file) check
Wrong mode selection	Use ios::in / ios::out correctly

Summary

- Always include `<fstream>` header.
- Use `ofstream` for writing, `ifstream` for reading, and `fstream` for both.
- Check file open success with `if (!file)`.
- Always close files after use.
- Modes control how data is handled (read, write, append, etc.).